

Monetary Policy and the Labor Market: A Quasi-Experiment in Sweden

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- Riksbank raised interest rates 2 p.p. in 2010-2011
- Credible, unanticipated deviation from historical policy rule
- Use quasi-experiment to study effects on labor market

- Background on Swedish economy
- Identification strategy
 - Tightening was unanticipated by market participants
 - Tightening was credible
 - Deviation from historical policy rule
- Aggregate effects of monetary policy shocks
- Heterogeneity in effects of monetary policy shocks

Background Information

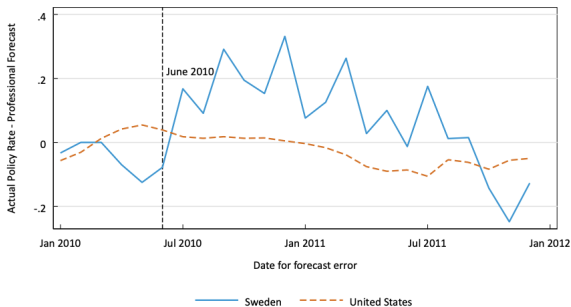
- Riksbank follows a policy of flexible inflation targeting
- Following 2008 financial crisis Riksbank cut rates to .25%
- By 2010, GDP grew by 6%, unemployment rate fell 1 p.p.
- Sweden was still recovering from the recession
 - GDP in 2010Q1 5% below pre-recession peak
 - Unemployment rate 2.5% above the natural rate

2010 Monetary Tightening

- Riksbank raised rates stepwise from .25% to 2%
- Not only reflecting positive assessment of current growth
- Riksbank members took into consideration financial conditions
 - Rising housing prices
 - Increase in household indebtedness

Tightening unanticipated by market participants

Figure 2: Professional Forecaster Expectations for the Policy Rate



- Forecasters consistently underestimated policy rate

Higher interest rate policy was credible

- Longer-term rates affected by movements in repo rate
- Only in 2012 did tensions within Riksbank become public

Deviation of policy rule

- Estimate following regression:

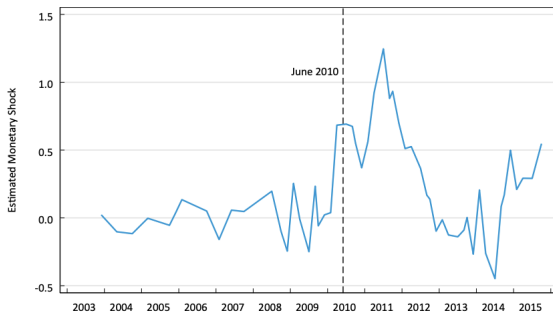
$$\Delta r_m = \alpha + \beta r_m + \sum_{\tau=-1}^2 \gamma_{\tau} GDP_{m,\tau} + \sum_{\tau=-1}^2 \phi_{\tau} \pi_{m,\tau} + \sum_{\tau=-1}^2 \theta_{\tau} u_{m,\tau} + \epsilon_m$$

- Define the monetary shock as the residuals:

$$\widehat{RR}_m = \epsilon_m$$

Deviation of policy rule

Figure 1: Estimated Monetary Policy Shocks



- Estimate regression on data from March 2002-February 2010

Summary of Identification Strategy

- Rate increases unexpected, credible deviation from policy rule
- Deviation caused by Riksbank reacting to financial conditions
- Estimate monetary policy shocks using RR methodology

Aggregate Effects of Monetary Policy

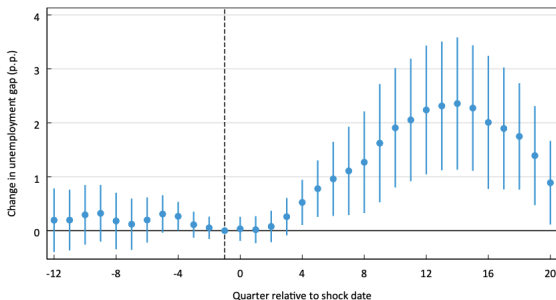
- Estimate a set of local projections

$$\tilde{u}_{t+k} - \tilde{u}_{t-1} = \beta_k \widehat{RR}_t + X'_{t-1} \alpha + \epsilon_t$$

- Zero out shocks not in 2010-2011 window
- Estimation sample includes all quarters from 1996Q1-2019Q2

Aggregate Effects of Monetary Policy

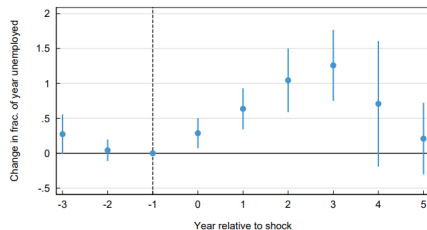
Figure 3: Effects of Monetary Policy on Aggregate Unemployment Rate



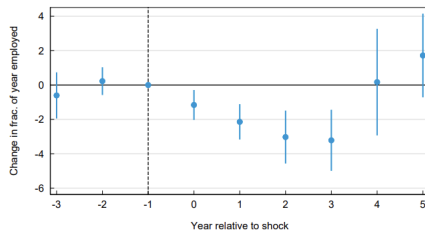
- Unemployment gap increases about 2 p.p.

Aggregate Effects of Monetary Policy: Micro Data

(a) Unemployment



(b) Employment



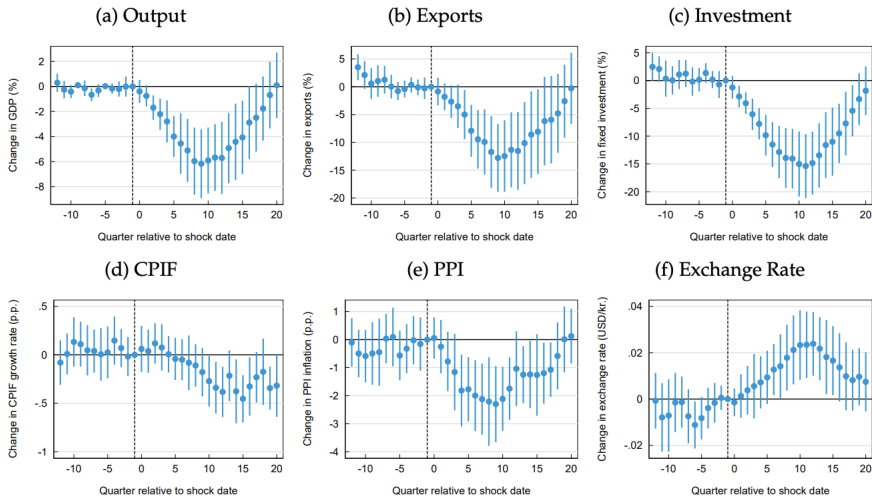
- Similar regression using micro data, individual-level controls

Other Estimates in Literature

- Estimates high compared to the literature
- RR methodology estimates rise of unemployment by 1 p.p.
- Only looking at contractionary shocks near Z.L.B.

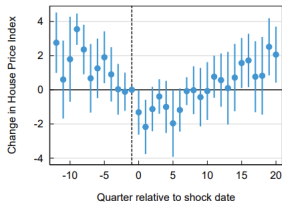
Effects on Other Aggregate Variables

Figure 6: Real Effects of Monetary Policy: Alternate Outcomes

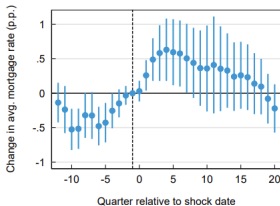


Effects on Other Aggregate Variables

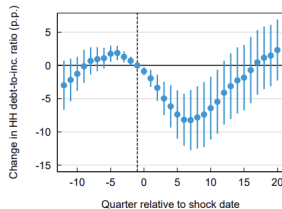
(g) House Price Index



(h) Average mortgage rate



(i) Household debt-to-income



- Have pre-effects, evidence of change in policy rule

Heterogeneity in Response to Shock

- Estimate effects using a set of local projections

$$u_{i,t+k} - u_{i,t-1} = \sum_{\omega} \left[\sum_{g \in \mathcal{G}_{\omega}} \left[\theta_{k,g} \widehat{RR}_t + X'_{t-1} \phi_{k,g} \right] \mathbb{1}(i \in g) \right] + Z'_i \gamma_k + \nu_{i,t}$$

- $\theta_{k,g}$ represents conditional k -year later response to shocks
- Estimating difference in effects between workers in group g and workers in a baseline group

Worker Heterogeneity

- Workers 25-34 spend .5% more of year unemployed than 55-64
- Low tenure workers spend 1.5% more of year unemployed
- Workers with HS degree spend .5% more of year unemployed

Firm Heterogeneity

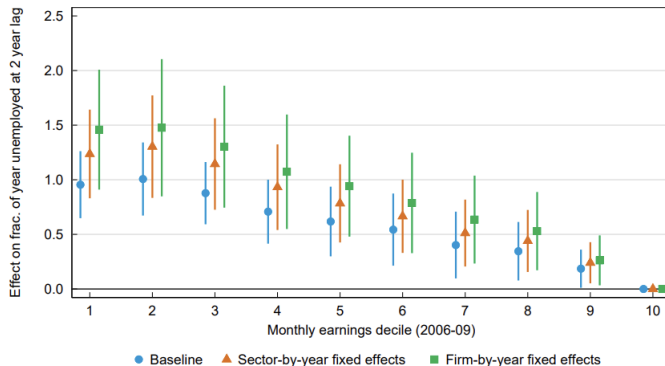
- Workers at indebted firms spend .5% more of year unemployed
- Workers at young firms spend .5% less of year unemployed
- Workers at small firms spend .3% more of year unemployed

Relative Importance of Firm and Worker Characteristics

- For each individual calculate predicted employment effect
- Worker characteristics main source of variance
- Worker tenure explains 85% of variance

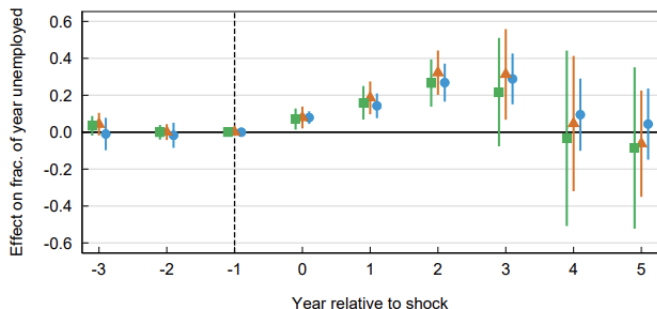
Effects on Inequality

Figure 9: Incidence Across the Earnings Distribution



Labor Market Rigidity

(a) Labor Market Contract

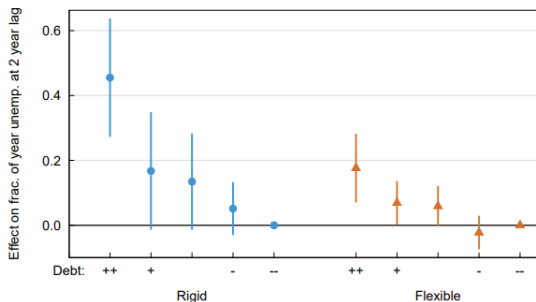


Difference for Rigid to Flexible: ■ Unconditional ▲ Conditional ● Conditional + Individual

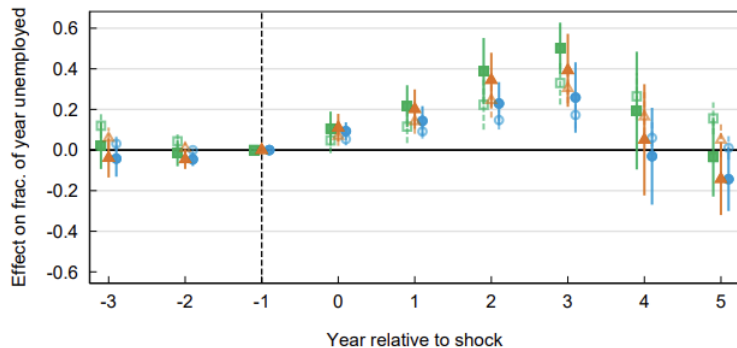
Labor Market Rigidity

- Wage rigidities affect unemployment through 2 channels
 - Congestion in labor market
 - Driving firm-level differences in exposure

(a) Firm Debt



(b) Churn



Difference for High to Low Churn ■ Unconditional ▲ Conditional ● Conditional + Individual

Conclusion

- Estimate 2 p.p. increase in unemployment
- Unemployment effects are heterogeneous among workers
 - Depend on worker and firm characteristics
 - Worker characteristics biggest drivers of differences
- Workers with more rigid wage contracts hit harder
 - Spend .3% more of year unemployed

Robustness of Labor Market Effects

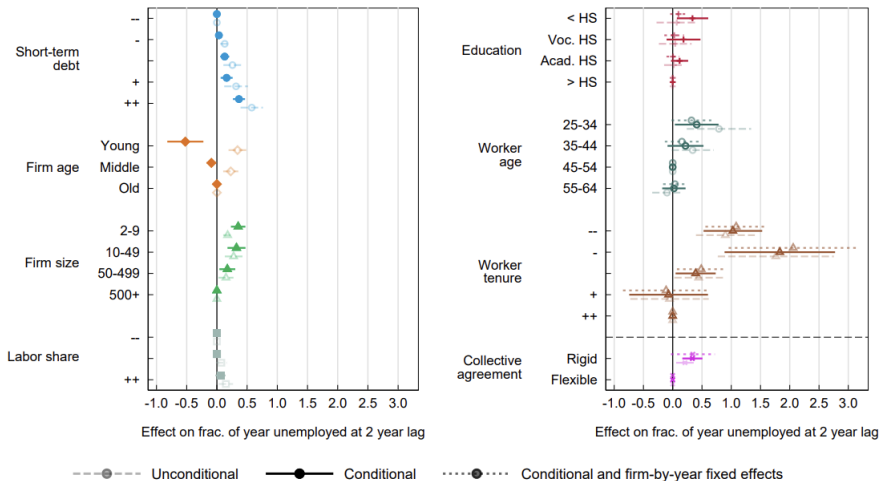
- Possible confounding effects from European debt crisis
- Add various controls
 - Export growth
 - Interbank rates for eurozone area
 - Average GDP of Sweden's trading partners
- Repeat estimates using workers at non-exporting firms

Robustness of Labor Market Effects

- Take event study approach
- Find results similar to using Romer and Romer shocks
- Run two different placebo tests
 - Randomly set dummy shocks to 1 in different periods
 - Run baseline specification in Finland
- Back

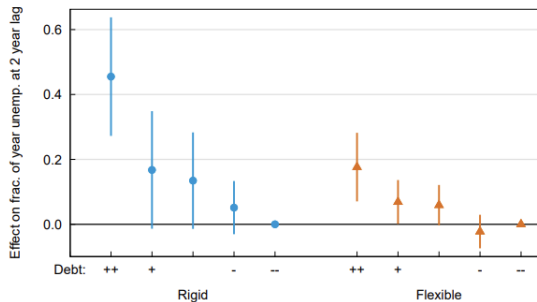
Firm and Worker Heterogeneity

Figure 8: Firm and Worker Heterogeneity in Exposure to Monetary Shock



Labor Market Rigidity

(a) Firm Debt



- Back